

NC State University
Department of Electrical and Computer Engineering
ECE 463/563 (Prof. Rotenberg)
Project #1: Cache Design, Memory Hierarchy Design
REPORT TEMPLATE (Version 2.0)

by

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NCSU Honor Pledge: "I have neither given nor received unauthorized aid on this project."

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Course number: 463

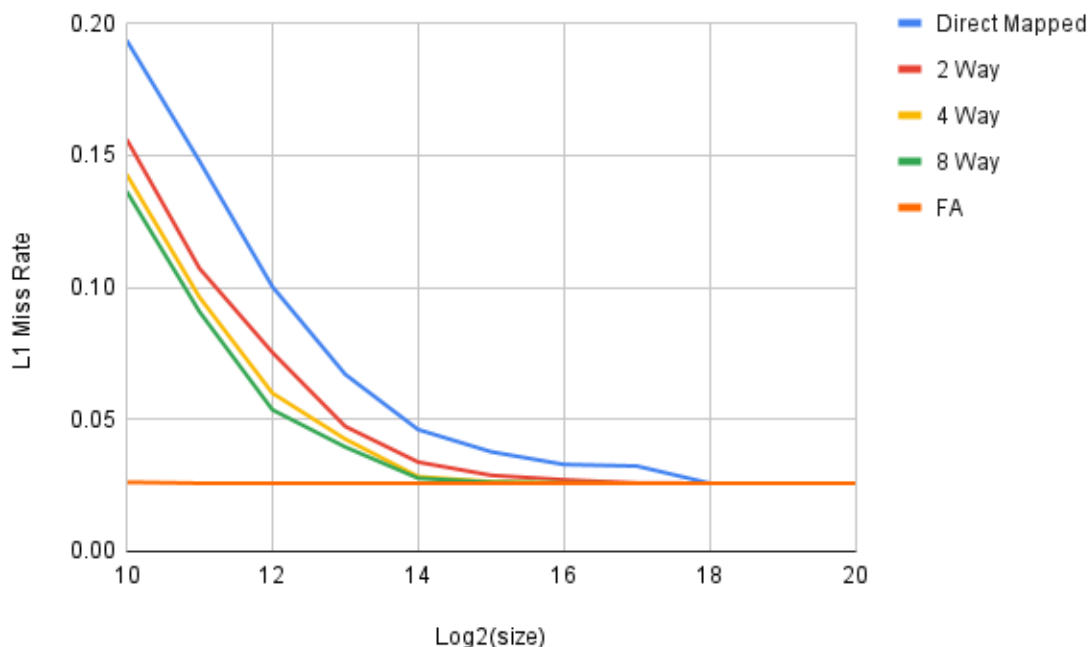
1. L1 cache exploration: SIZE and ASSOC

GRAPH #1 (total number of simulations: 55)

For this experiment:

- Benchmark trace: gcc_trace.txt
- L1 cache: SIZE is varied, ASSOC is varied, BLOCKSIZE = 32.
- L2 cache: None.
- Prefetching: None.

Plot L1 miss rate on the y-axis versus $\log_2(\text{SIZE})$ on the x-axis, for eleven different cache sizes: SIZE = 1KB, 2KB, ..., 1MB, in powers-of-two. (That is, $\log_2(\text{SIZE}) = 10, 11, \dots, 20$.) The graph should contain five separate curves (*i.e.*, lines connecting points), one for each of the following associativities: direct-mapped, 2-way set-associative, 4-way set-associative, 8-way set-associative, and fully-associative. All points for direct-mapped caches should be connected with a line, all points for 2-way set-associative caches should be connected with a line, *etc.*



Answer the following questions:

1. For a given associativity, how does increasing cache size affect miss rate?

Increasing the size lowers the miss rate

2. For a given cache size, how does increasing associativity affect miss rate?

Increasing associativity lowers miss rate

3. Estimate the *compulsory miss rate* from the graph and briefly explain how you arrived at this estimate.

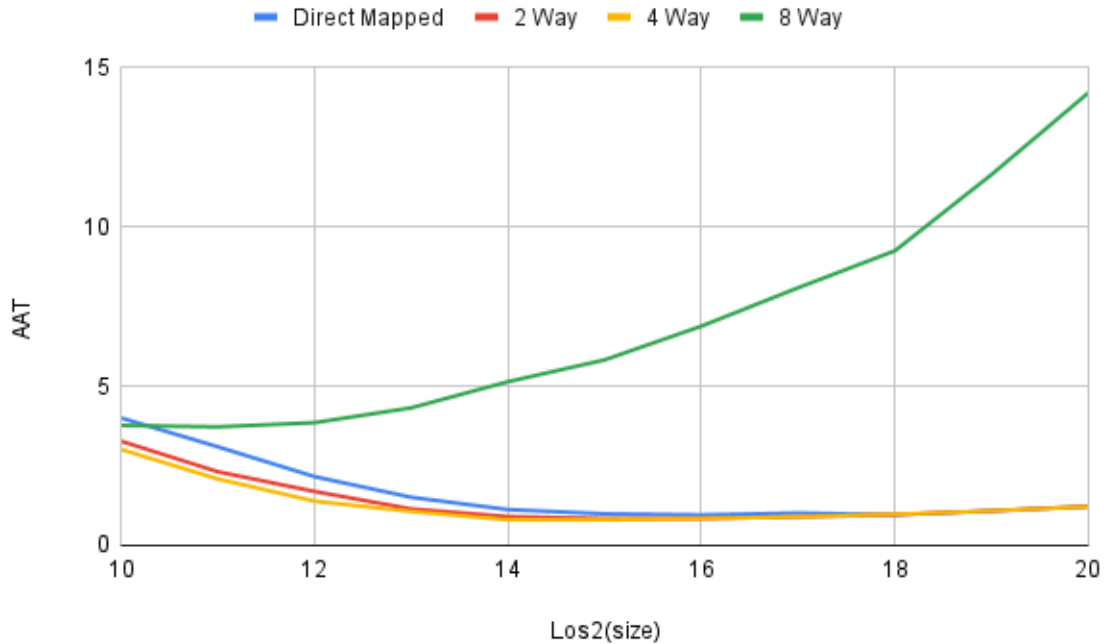
compulsory miss rate = 0.0258

How I arrived at this estimate: From the graph we can deduce the compulsory rate when just looking at a fixed associativity and fixed block size. Also in fully associative the floor for miss rates is 0.0258.

GRAPH #2 (no additional simulations with respect to GRAPH #1)

Same as GRAPH #1, except make the following changes:

- The y-axis should be AAT instead of L1 miss rate.
- Do NOT include fully-associative cache configurations (power-hungry, impractical cost for implementing LRU). Thus, GRAPH #2 should contain four (not five) separate curves: direct-mapped, 2-way set-associative, 4-way set-associative, 8-way set-associative.



Answer the following question:

1. For a memory hierarchy with only an L1 cache and BLOCKSIZE = 32, which configuration yields the best (*i.e.*, lowest) AAT and what is that AAT?

Direct-mapped or set-associative cache configuration that yields the lowest AAT:

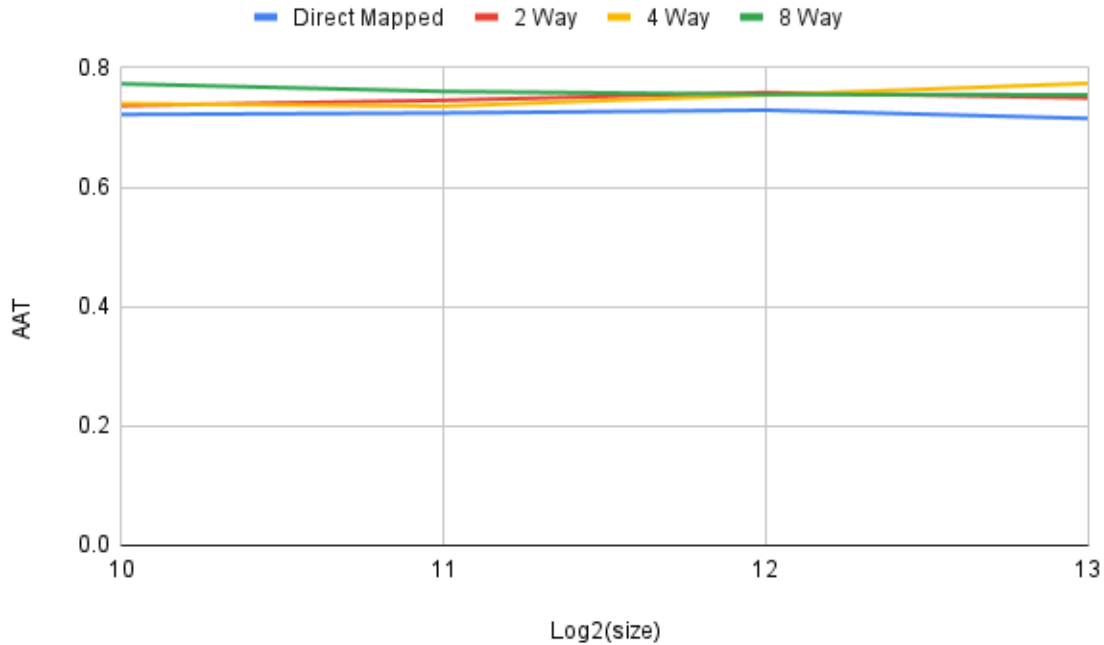
32 KB, 4-way set-assoc.

AAT for this configuration: 0.80189

GRAPH #3 (total number of simulations: 16)

Same as GRAPH #2, except make the following changes:

- Add the following L2 cache to the memory hierarchy: 16KB, 8-way set-associative, same block size as L1 cache.
- Vary the L1 cache size only between 1KB and 8KB (since L2 cache is 16KB). (And as with GRAPH #2, GRAPH #3 should contain four (not five) separate curves for L1 associativity: direct-mapped, 2-way set-associative, 4-way set-associative, 8-way set-associative.)



Answer the following questions:

1. With the L2 cache added to the system, which L1 cache configuration yields the best (*i.e.*, lowest) AAT and what is that AAT?

L1 configuration that yields the lowest AAT with 16KB 8-way L2 added:

8 KB, Direct-Mapped.

AAT for this configuration: **0.715107608**

2. How does the lowest AAT with L2 cache (GRAPH #3) compare with the lowest AAT without L2 cache (GRAPH #2)?

The lowest AAT with L2 cache is **less than** the lowest AAT without L2 cache.

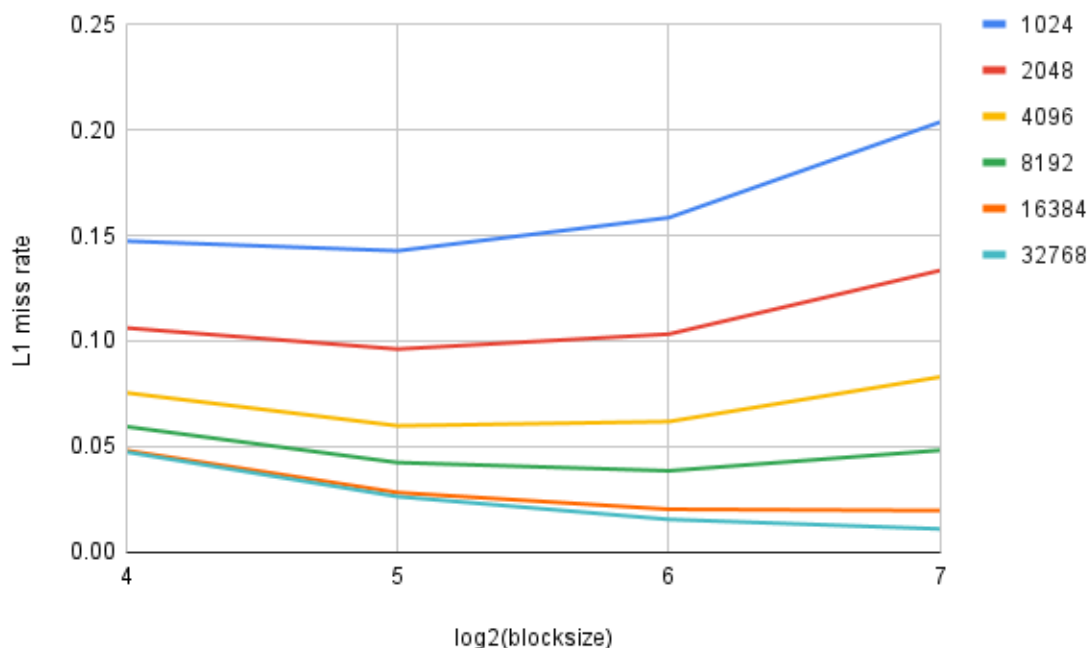
2. L1 cache exploration: SIZE and BLOCKSIZE

GRAPH #4 (total number of simulations: 24)

For this experiment:

- Benchmark trace: gcc_trace.txt
- L1 cache: SIZE is varied, BLOCKSIZE is varied, ASSOC = 4.
- L2 cache: None.
- Prefetching: None

Plot L1 miss rate on the y-axis versus $\log_2(\text{BLOCKSIZE})$ on the x-axis, for four different block sizes: BLOCKSIZE = 16, 32, 64, and 128. (That is, $\log_2(\text{BLOCKSIZE}) = 4, 5, 6,$ and 7.) The graph should contain six separate curves (*i.e.*, lines connecting points), one for each of the following L1 cache sizes: SIZE = 1KB, 2KB, ..., 32KB, in powers-of-two. All points for SIZE = 1KB should be connected with a line, all points for SIZE = 2KB should be connected with a line, *etc.*



Answer the following questions:

1. Do smaller caches prefer smaller or larger block sizes?

Smaller caches prefer **smaller** block sizes. For example, the smallest cache considered in Graph #4 (1KB) achieves its lowest miss rate with a block size of **32 B**.

2. Do larger caches prefer smaller or larger block sizes?

Larger caches prefer **larger** block sizes. For example, the largest cache considered in Graph #4 (32KB) achieves its lowest miss rate with a block size of **128 B**.

3. As block size is increased from 16 to 128, is the tension between *exploiting more spatial locality* and *cache pollution* evident in the graph? Explain.

Yes, the tension between *exploiting more spatial locality* and *cache pollution* is evident in the graph.

For example, consider the smallest (1KB) cache in Graph #4. Increasing block size from 16 B to 32 B is helpful (reduces miss rate) due to *exploiting more spatial locality*. But then increasing block size further, from 32 B to 128 B, is not helpful (increases miss rate) due to *cache pollution* having greater effect.

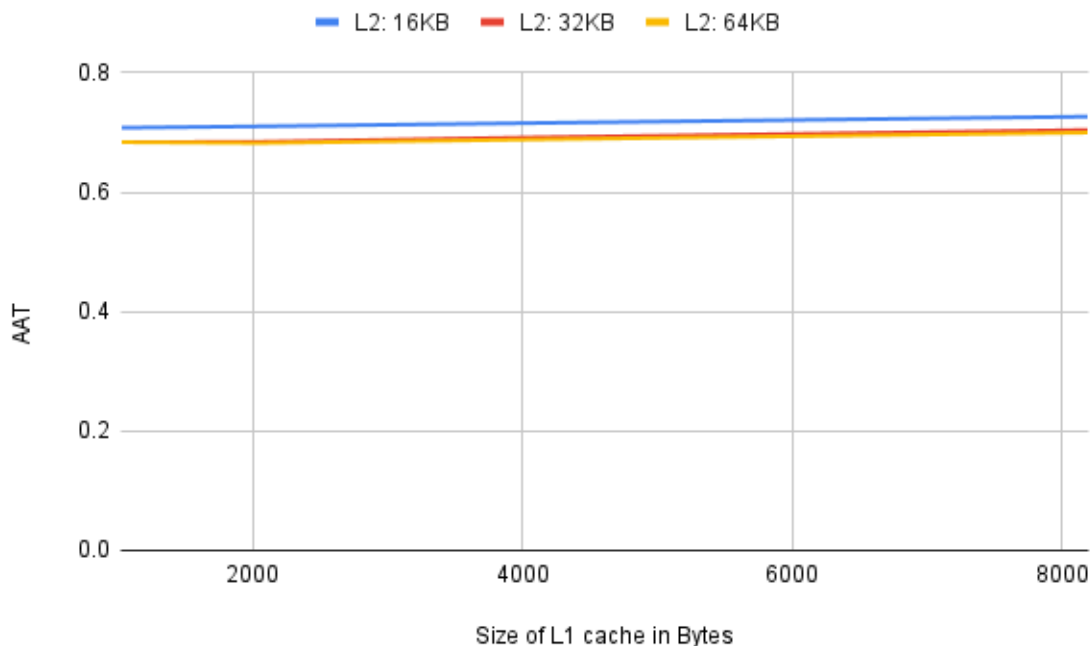
3. L1 + L2 co-exploration

GRAPH #5 (total number of simulations: 12)

For this experiment:

- Benchmark trace: gcc_trace.txt
- L1 cache: SIZE is varied, BLOCKSIZE = 32, ASSOC = 4.
- L2 cache: SIZE is varied, BLOCKSIZE = 32, ASSOC = 8.
- Prefetching: None.

Plot AAT on the y-axis versus $\log_2(\text{L1 SIZE})$ on the x-axis, for four different L1 cache sizes: L1 SIZE = 1KB, 2KB, 4KB, 8KB. (That is, $\log_2(\text{L1 SIZE}) = 10, 11, 12, 13$.) The graph should contain three separate curves (*i.e.*, lines connecting points), one for each of the following L2 cache sizes: 16KB, 32KB, 64KB. All points for the 16KB L2 cache should be connected with a line, all points for the 32KB L2 cache should be connected with a line, *etc.*



Answer the following question:

1. Which memory hierarchy configuration in Graph #5 yields the best (*i.e.*, lowest) AAT and what is that AAT?

Configuration that yields the lowest AAT: L1: 8KB 4 Way L2: 64KB 8 Way
Lowest AAT: 0.6822518326

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